

Beyond the Empathy Fallacy: A Meta-analysis of Artificial Empathy in Human-AI Interactions

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Abstract

Artificial empathy is widely adopted in conversational agents, which is grounded in the prevailing belief that emotionally expressive agents foster trust and engagement. While generative language models enable increasingly fluent and human-like empathic cues, it remains unclear whether artificial empathy consistently benefits users or instead produces unintended consequences for trust and behavioral engagement. We conducted a meta-analysis of 36 controlled studies drawn from 23 articles to examine how different dimensions of artificial empathy shape user responses across task contexts and technological eras. Results reveal a clear asymmetry: while cognitive empathy reliably supports behavioral engagement, affective empathy produces heterogeneous effects and can undermine behavioral intentions, particularly in emotional support tasks and within the Generative AI era. These findings challenge the assumption that greater empathy is inherently beneficial and provide an evidence-based framework for the strategic design of empathic AI.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; *HCI theory, concepts and models*.

Keywords

Artificial Empathy, Meta-Analysis, Conversational Agents, Human-AI Interaction

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1 Introduction

In Human-Computer Interaction (HCI), artificial empathy has increasingly been treated as a normative design goal, grounded in the belief that making AI agents emotionally expressive will bring human-AI interaction closer to human social dynamics [2, 27, 28]. This orientation reflects an implicit *empathy fallacy*: the assumption that progress in AI interaction design is best achieved by approximating human emotional experience as closely as possible [1]. Recent advances in Generative AI bring this assumption into sharper focus [5]. Large language models can now generate responses with unprecedented fluency, producing highly realistic simulations of empathic concern and emotional attunement [33]. This technical leap raises a critical empirical question for HCI: Do users actually want empathic AI that appears to “feel”, or does intensified emotional simulation instead undermine trust and engagement?

Prior research on artificial empathy commonly distinguishes between cognitive and affective dimensions [11, 20]. Cognitive empathy emphasizes situational appraisal and perspective-taking, aligning with users’ expectations of AI as an information-processing agent and often yielding robust improvements in perceived competence [35]. In contrast, affective empathy relies on expressions that simulate emotional resonance and has been associated with more heterogeneous user responses [28]. Empirical evidence suggests that such affective cues may conflict with users’ role expectations of artificial agents, triggering expectancy violations that undermine perceived authenticity and continued use [3, 26, 36]. Moreover, the effectiveness of artificial empathy is highly contingent on task context. Empathic cues may be welcomed in some emotional support settings [12, 19], yet become less effective or even counterproductive in instrumental or efficiency-oriented tasks [14, 23]. Consequently, these findings point to substantial heterogeneity in how artificial empathy shapes user responses, highlighting the need for a systematic synthesis of how empathy dimensions and task context jointly condition its effects.

This study adopts a meta-analytical design to examine how different dimensions of artificial empathy function across task contexts and technological eras. Instead of treating artificial empathy as a unitary construct, we quantify the distinct effects of cognitive and affective empathic cues on users’ affective responses, such as emotion regulation, and on downstream behavioral outcomes, including continued usage intentions. We further analyze how the transition



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to the Generative AI era reshapes the relationship between simulated empathy and user responses, with particular attention to the moderating role of task type, such as emotional support versus instrumental interaction. By synthesizing empirical evidence across this body of research, we challenge the prevailing assumption that increasing empathic expressiveness is inherently beneficial and provide an evidence-based framework for the strategic design of empathic agents.

2 Methods

A meta-analysis provides a systematic way to synthesize evidence from prior studies that examine different interventions and report heterogeneous outcome measures [8, 31]. This approach enables structured aggregation of findings across studies that vary in context and design. Following the PRISMA guidelines, we conducted a systematic process to identify, screen, and synthesize relevant studies.

2.1 Data Collection

We employed three complementary search strategies. First, we conducted systematic searches across four major databases: ACM Digital Library, Scopus, Web of Science, and EBSCO, combining AI agent-related terms (e.g., chatbot, conversational agent, LLM) with empathy-related keywords (e.g., empathy, empathic, empathetic, compassionate). Second, we performed backward and forward citation searches and manually reviewed leading journals and conference proceedings. Third, we included gray literature such as dissertations and preprints to mitigate potential publication bias. This process yielded 2,012 records.

Screening criteria were then applied to identify eligible studies that examined the effects of artificial empathy cues in human-AI interaction. Studies were included if they met the following criteria: (1) the use of an experimental design with clearly identifiable manipulations and explicit comparison conditions between AI agents with and without empathy cues (e.g., empathic vs. non-empathic language); (2) the investigation of interactive conversational AI agents, such as chatbots or voice assistants; and (3) the reporting of quantitative outcome measures capturing users' affective responses (e.g., relational perceptions or emotional states) or behavioral intentions (e.g., continued usage intention). Studies were excluded if they (1) adopted non-experimental designs; (2) focused on static or one-way interaction formats; (3) lacked sufficient statistical information for effect size computation. The final dataset comprised 23 articles representing 36 independent studies published between 2018 and 2025 [4, 7, 9, 12–15, 17, 18, 20–26, 29, 30, 32, 34, 39, 41, 42].

Two independent annotators coded 30% of the effect estimates to assess coding reliability. Inter-coder agreement across all coded dimensions exceeded the recommended threshold (Krippendorff's $\alpha > 0.80$). Remaining discrepancies were resolved through discussion, after which a single annotator completed the remaining coding. Given the heterogeneous operationalizations of artificial empathy across studies, we harmonized empathy expressions using a unified theoretical framework to enable systematic comparison. This classification framework draws on established theoretical distinctions in empathy research [6, 11], thereby providing a conceptually

grounded basis for harmonization rather than post-hoc reclassification. Empathy-related experimental materials and descriptive texts were first extracted from each study and then classified using a few-shot LLM-based procedure (GPT-4o), guided by predefined theoretical criteria summarized in Table 1. Multiple labels per study were allowed to capture complex and hybrid empathic designs. To validate the LLM-assisted classification, a human expert independently coded a random subset of the materials. Agreement between human and LLM classifications was high across empathy categories (average Krippendorff's $\alpha = 0.853$), indicating strong consistency between automated and expert judgments.

Table 1: Operationalization of Empathy Dimensions with Representative Examples

Empathy Dimension	Operationalization	Representative Examples
Cognitive Empathy	Coded 1 when the agent demonstrated situational appraisal or intellectual recognition of the user's state without expressing simulated affect.	• "I understand your anxiety"[29] • "I understand that you may be a little impatient now, but please give me another chance to serve you." [23]
Affective Empathy	Coded 1 when the agent expressed emotional responsiveness through simulated resonance or affective self-disclosure.	• "I could feel your anxiety about being noticed by others." [41] • "I wish I could do more." [25]

2.2 Data Analysis

To ensure comparability across studies with different designs and sample sizes, Hedges' g was used as the primary effect size metric. Hedges' g is a standardized mean difference that applies a correction for small-sample bias and therefore provides more robust and unbiased estimates than Cohen's d [16]. Effect sizes were primarily calculated from reported means, standard deviations, and sample sizes for experimental and control groups. When these statistics were unavailable, effect sizes were derived from alternative reported statistics (e.g., t values, F values, or correlation coefficients) and converted into Hedges' g using established formulas. All effect sizes were computed in R.

Pooled effect sizes were first estimated using random-effects models and visualized with forest plots. To account for the nested structure of effect sizes within studies and articles, a three-level mixed-effects meta-analytic framework was adopted [10]. This approach partitions variance across hierarchical levels and reduces bias arising from non-independent observations. Within this framework, mixed-effects meta-regressions were conducted separately for affective responses and behavioral intentions [31]. Empathy dimensions and task context were included as primary moderators, with interaction terms added to examine how empathic design effects vary across technological eras (Legacy vs. GenAI).

3 Results

3.1 Overall Effects and Heterogeneity

Figure 1 presents the forest plots for affective responses and behavioral intentions. For affective responses, most studies reported positive effect sizes, with several estimates of moderate to large magnitude. In contrast, effect sizes for behavioral intentions were more dispersed, including both positive and negative estimates.

Results from three-level mixed-effects models indicated that artificial empathy was positively associated with affective responses ($g = 0.45$, 95% CI [0.29, 0.61]) and behavioral intentions ($g = 0.34$,

95% CI [0.18, 0.51]). Both outcomes exhibited substantial heterogeneity ($Q_{aff}(22) = 110.92, p < .0001$; $Q_{beh}(30) = 514.66, p < .0001$; $I^2 > 86\%$), supporting the adoption of our model and subsequent moderator analyses. Trim-and-fill analyses suggested that publication bias does not fundamentally alter the interpretation of the results.

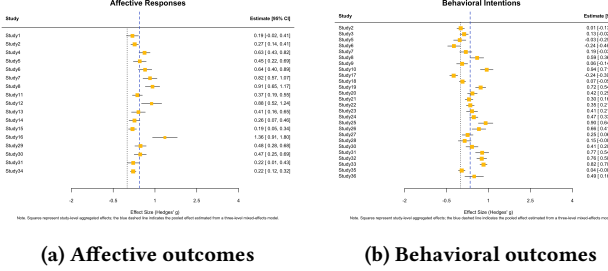


Figure 1: Forest plots of the main effects of artificial empathy.

3.2 Divergent Effects of Empathy Dimensions

To examine the distinct impact of each empathy dimension, we estimated base models for cognitive and affective empathy independently. These models estimated the main effects of empathy dimensions on user outcomes, as presented in Table 2. Behavioral intentions showed clearly differentiated patterns across empathy types. Cognitive empathy was positively associated with behavioral intentions ($\beta_{Base} = 0.394, p < .001$), whereas affective empathy exhibited a significant negative association ($\beta_{Base} = -0.354, p = .002$). Affective responses did not show significant main effects in the base models. Neither affective empathy ($\beta_{Base} = 0.144, p = .217$) nor cognitive empathy ($\beta_{Base} = 0.030, p = .842$) reached statistical significance.

3.3 Task Context as a Boundary Condition

As summarized in Table 2, the interaction models examined whether task context moderated the effects of empathy type. Compared with the base models, interaction models exhibited improved fit, as indicated by consistently lower AIC values. Variance estimates further suggested a reduction in between-article heterogeneity after including task context as a moderator. Within emotional support tasks, the negative effect of affective empathy was amplified ($\beta_{Int} = -0.694, p = .002$), while the positive effect of cognitive empathy remained significant ($\beta_{Int} = 0.476, p = .024$).

Moderation effects on affective responses were weaker. The interaction between cognitive empathy and emotional support tasks showed a marginally significant positive trend ($\beta_{Int} = 0.586, p = .066$).

3.4 GenAI Era as an Amplifier

To capture the impact of the recent transition to Generative AI, we categorized studies into *Legacy AI* (pre-2023) and *GenAI* (post-2023). The interaction models revealed a differentiated pattern across outcomes, as visualized in Figure 2. For affective responses, a significant interaction was observed between cognitive empathy and technological era ($\beta = 0.861, p = .027$), indicating that cognitive empathy

Table 2: Comparison of Base and Interaction Models: Impact of Empathy Type and Task Context.

Outcome Category	Empathy Type	Base Model ¹			Interaction Model ²		
		β_{Base}	p_{Base}	AIC _B	$\beta_{Int} (\times \text{Task Type})$	p_{Int}	AIC _I
Affective Responses	Affective	0.144	.217	9.53	0.078	.755	13.68
	Cognitive	0.03	.842	10.46	0.586 [†]	.066	11.34
Behavioral Intentions	Affective	-0.354**	.002	129.19	-0.694**	.002	123.97
	Cognitive	0.394***	<.001	124.35	0.476*	.024	122.98

Note: $k_{Aff} = 21, k_{Beh} = 31$. Task type was binary-coded into two categories: *emotional support* (coded 1) and *instrumental tasks* (coded 0; e.g., informational interactions). ¹ Base models estimate the main effects of empathy type on each outcome. ² Interaction models examine whether empathy effects vary across task contexts. [†] $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$.

was more strongly associated with affective responses in the GenAI era than in the legacy era. Conversely, for behavioral intentions, a significant negative interaction was observed for affective empathy ($\beta = -0.604, p = .042$), suggesting that the negative association between affective empathy and behavioral intentions was amplified in the GenAI era.

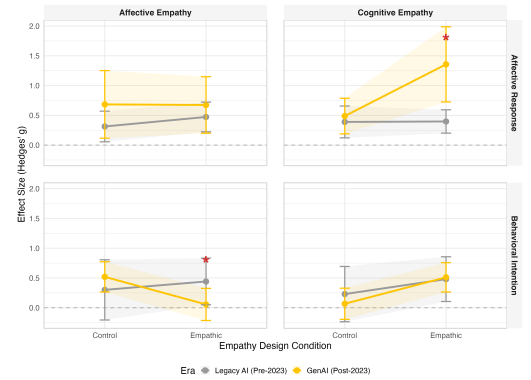


Figure 2: Moderation effects of the technological era (Legacy vs. GenAI) on the relationship between empathy type and user outcomes

4 Discussion

4.1 The Behavioral Backfire of Affective Empathy

Our meta-analysis reveals a systematic *Feeling–Doing Dissociation* in human–AI interaction. Although affective empathy is intended to convey warmth, it shows a consistent negative association with behavioral intentions, an effect amplified in the GenAI era ($\beta = -0.604, p = .042$). This pattern indicates that emotionally expressive support does not reliably translate into behavioral engagement and may, under some conditions, produce unfavorable behavioral responses. Prior studies suggest a similar divergence. Emotionally expressive agents can enhance affective responses such as enjoyment [19], yet such emotional comfort does not necessarily sustain continued use. Other work shows that affective empathy

can simultaneously increase warmth while lowering perceived authenticity or evoking eeriness, thereby discouraging engagement [35, 41]. One interpretation is that when users recognize AI as non-sentient, simulated emotional resonance may be evaluated as stylistic rather than experiential. Consequently, intensifying affective empathy can paradoxically undermine the perceived authenticity required for long-term behavioral engagement.

4.2 The Cognitive Dividend of Cognitive Empathy

In contrast, cognitive empathy demonstrates a consistent positive association with behavioral intentions, indicating that accurate situational understanding reliably translates to user engagement. This advantage is amplified in the GenAI era, where cognitive empathy is more strongly associated with affective responses ($\beta = 0.861, p = .027$). This pattern suggests a possible functional pathway: perspective-taking and situational awareness may serve as evidence of competence and understanding. Earlier concerns that cognitively empathic responses might appear emotionally cold [20] hold less weight in contemporary systems. With advances in generative AI, high-fidelity interpretation of users' goals and constraints may increasingly be perceived as credible understanding rather than mechanical processing [37, 40]. Compared with affective mimicry, cognitive empathy therefore provides a more stable foundation for both evaluative and behavioral responses. Accordingly, designers might increasingly prioritize demonstrating comprehension over simulating emotional resonance, leveraging generative fluency to explicitly validate users' situational needs and constraints.

4.3 Contextual Risks in Emotional Support Tasks

Task context further conditions these effects. In emotional support settings, the negative association between affective empathy and behavioral intentions is stronger ($\beta = -0.694, p = .002$). This reveals that artificial empathy requires precise calibration to the specific emotional vulnerabilities of the interaction. When users are vulnerable, they may become more attentive to whether empathic responses appear grounded rather than merely well-phrased. In such contexts, highly polished affective language may heighten awareness of the agent's artificiality, reinforcing the heuristic that chatbots lack genuine emotional capacity [26, 38]. Consequently, cognitive empathy may represent a more robust strategy in emotionally sensitive interactions, whereas affective displays introduce context-dependent risks. In these settings, minimizing generic or scripted affective language may help reduce skepticism and support more credible interaction experiences. These patterns highlight the importance of aligning empathic expression with contextual expectations. Future work should examine specific task characteristics, such as emotional valence and time pressure, as well as user-level factors such as vulnerability and cultural background, to advance adaptive empathic design in conversational agents.

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A Validation of LLM-based Empathy Classification

To evaluate the robustness of our few-shot classification approach, we compared the LLM-generated labels against human annotations. Table 3 present the classification reports for cognitive and affective empathy, respectively. The high accuracy and F_1 scores

demonstrate that the model is a reliable tool for large-scale empathy categorization in this meta-analysis.

Table 3: Predictive Performances of GPT-4o-based Empathy Classifiers

	Precision	Recall	F_1 score	Support
Cognitive Empathy				
No (0)	0.96	0.96	0.96	23
Yes (1)	0.96	0.96	0.96	28
Macro Avg	0.96	0.96	0.96	51
Accuracy			0.96	51
Affective Empathy				
No (0)	0.96	0.85	0.90	27
Yes (1)	0.86	0.96	0.91	25
Macro Avg	0.91	0.91	0.90	52
Accuracy			0.90	52

B Publication Bias Assessment

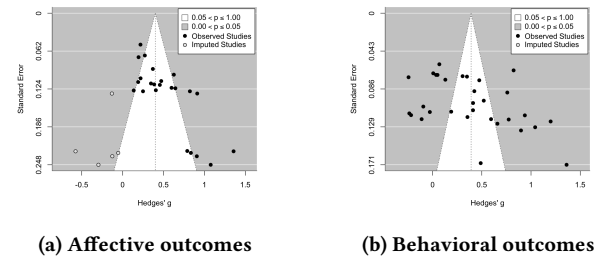


Figure 3: Trim-and-fill funnel plots for affective responses and behavioral intentions.